

Teja Ala

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🌐 [dekeract01.github.io](https://github.com/dekeract01)

🐙 [dekeract01](https://github.com/dekeract01)

EDUCATION

PhD, Aeronautical Engineering, *University of Southampton* Oct 2021 – Jan 2026

Thesis: *Laminar-Turbulent Transitions in High-Enthalpy Boundary Layers*

MEng (Hons) Mechanical Engineering, *University of Liverpool* Sep 2015 – Jun 2020

First-class honours: IMechE Best Project Award, Santander Team Award, Excellence Scholarship

Masters Thesis: *Elastic Turbulence as an Initial Value Problem*

The Blue Coat School Liverpool Sep 2013 – Jun 2015

A-levels: Maths (A*), Further Maths (A), Physics (A), Chemistry (A)

EMPLOYMENT

Modelling and Simulations Engineer , <i>Aerospace Company</i>	May 2026 – Present
Teaching Assistant , <i>University of Southampton</i>	Nov 2021 – Dec 2024
Research and Teaching Assistant , <i>University of Liverpool</i>	Jul 2020 – Aug 2021
MATLAB Ambassador , <i>MathWorks</i>	Sep 2019 – Aug 2020
Mechanical Analysis Intern , <i>Airbus Defence and Space</i>	Aug 2018 – Aug 2019
Software Placement , <i>The Institute for Risk and Uncertainty</i>	Aug 2018 – Aug 2019
Sales Assistant , <i>Screwfix</i>	Aug 2013 – Aug 2016

VOLUNTEERING & INTERESTS

Postgraduate Seminar Organiser , <i>University of Southampton</i>	Jan 2022 – Nov 2024
Publicity Manager , <i>Royal Aeronautical Society Solent</i>	2024
Postgraduate Ambassador , <i>University of Southampton</i>	2022
Organiser , <i>HiPy Python, Liverpool</i>	2020
Executive Member , <i>Royal Aeronautical Society Stevenage</i>	2019
Ambassador , <i>STEM and IET Stevenage</i>	2018
Balance for Business , <i>Airbus</i>	2018
Student Ambassador/Chair , <i>Staff-Student Liaison Committee, Liverpool</i>	2017
Team Leader , <i>Liverpool Karting Team</i>	2017
Secretary , <i>Liverpool Flight Simulator Group</i>	2017

CONFERENCES & TALKS

UK Hypersonics SIG , <i>Bristol</i> – Presentation	2026
UK Fluids , <i>Liverpool</i> – Presentation	2025
AIAA SciTech , <i>Orlando, Florida</i> – Paper presentation	2025
OpenSBLI Workshop , <i>Oxford</i> – Presentation	2023
RAeS Hypersonics Conference , <i>London</i> – Presentation	2023
Southampton Aerospace & Astrodynamics Conference , <i>Southampton</i> – Presentation	2022
UK Fluids , <i>Sheffield</i> – Poster	2022

PUBLICATIONS

Ala, T. et al., <i>Extension of OpenSBLI v4.0 to Thermochemical Nonequilibrium</i> , In preparation	2026
Ala, T. and Sandham, N.D., <i>Breakdown of Hypersonic Boundary-Layer Disturbances under Thermal Nonequilibrium</i> , AIAA Journal	2026 (Preprint)
Ala, T. , <i>Laminar-Turbulent Transition in High-Enthalpy Boundary Layers</i> , Doctoral Thesis, University of Southampton	2026
Ala, T. and Sandham, N.D., <i>Effect of Thermal Nonequilibrium on Disturbance Growth in Hypersonic Boundary Layers</i> , AIAA SciTech	Jan 2025

SKILLS

- **Languages:** Python, SQL, MATLAB, C/C++, Fortran, Bash
- **Analytics & ML:** NumPy, pandas, SciPy, scikit-learn, PyTorch, HDF5
- **CFD:** OpenSBLI, OpenFOAM, SU2, CFD++, ANSYS Fluent, StarCCM+
- **CAD & Meshing:** Creo, SolidWorks, Fusion 360, Pointwise, Gmsh
- **HPC:** Linux, Slurm, MPI, OpenMP, CUDA, OpenCL, OpenACC, mpi4py, Numba
- **Visualisation:** ParaView, Tecplot, Matplotlib, Seaborn
- **Tools:** Git, Docker, CMake, Jupyter